An Automated Computer Vision System for Bagged Cement Product Inspection and Counting for Stock Management

ระบบตรวจสอบและตรวจนับผลิตภัณฑ์ปูนซีเมนต์บรรจุถุงแบบอัตโนมัติด้วยคอมพิวเตอร์วิทัศน์สำหรับกระบวนการบริหารคลังสินค้า

1. Project Background & Understanding

Under the current operational framework, cement bag throughput is measured using a Relay Counter; however, the identification of stock keeping units (SKUs) remains dependent on manual input by plant operators within the platform interface. This disconnect between physical packaging, and system selection introduces human error, leading to discrepancies between recorded inventory and actual dispatch. Furthermore, quality assurance relies on post-production manual inspections, which delays the detection of packaging breaches and cement leakage.

To resolve these limitations, this project will deploy an automated Computer Vision and Image Processing system across production lines. The solution transitions the operational workflow from an unverified, manual process (Operator-defined SKU → Relay Counter → Assumed SKU Count) to a fully automated architecture (Camera → Actual SKU Recognition → Bag Detection → Count → Production Data). Cameras installed on the line will automatically classify bag SKUs, track real-time production volumes, and flag severe packaging defects. This system provides reliable real-time production data as the foundation for real-time inventory visibility and future slot-booking capabilities.

System acceptance and operational performance will be evaluated against four primary Key Performance Indicators (KPIs):

1. SKU Recognition: Accuracy ≥ 98\%
2. Bag Counting: Accuracy ≥ 98\%
3. Defect Detection: Recall ≥ 98\%
4. Processing Latency: End-to-end latency ≤ 1 minute

The solution architecture is designed to support long-term scalability across more than 10 factory locations and approximately 40 distinct SKUs. The hardware and software components must operate reliably within high-speed production environments and high-dust conditions typical of cement manufacturing. To ensure accurate accounting, the system will maintain real-time tracking across three distinct metrics: Gross Count, Defect Count, and Good Count, ensuring defective units are logged separately and excluded from the Good Count total. Defect categorization is restricted to high-severity failures (large tears and visible leakage), and uncaptured defects on the bottom of bags are recognized as a hardware placement limitation.

2. Proposed Solution

The solution is delivered in two phases: a 3-month Proof of Concept (PoC) to validate model accuracy and system logic, followed by a 7-month production rollout that integrates the models into the company's existing platform and deploys them across all factories.

Phase 1: Proof of Concept (Months 1-3)
Operational cameras will be installed to capture 7 days of continuous production footage for dataset annotation. Computer vision models will be trained to recognize approximately 40 SKUs and detect severe defects in alignment with the target KPIs. Production control logic will incorporate a configurable production-idle timeout to halt counting during line inactivity, alongside automated SKU transition detection. Technical evaluations will compare timestamped queue-based processing, edge computing, and cloud infrastructure (e.g., AWS or GCP) to determine the optimal processing framework to support line speed requirements.

Phase 2: Production Integration & Rollout (Months 4-10)
The validated models will be integrated into the company's existing TypeScript-based platform and connected via AWS IoT. A self-service artwork upload feature will be implemented, allowing operational teams to onboard new bag packaging designs without requiring ongoing software development. System deployment will follow a phased approach, beginning with pilot plant validation before expanding across 10+ factory locations. The phase concludes with User Acceptance Testing (UAT), administrative handovers, and staff training.

3. Key Features & Deliverables

1. **Real-Time SKU Recognition & Production Counting:** Automated SKU identification coupled with triple-metric volume tracking (Gross Count, Defect Count, and Good Count).
2. **Automated Production Logic:** Intelligent line control featuring a configurable production-idle timeout and automatic SKU switch detection.
3. **Defect Detection Module:** Real-time identification of severe structural damage and cement leakage, excluding compromised bags from net production totals.
4. **Self-Service SKU & Artwork Management:** An operational upload tool enabling users to register new packaging artwork seamlessly.
5. **Enterprise Platform & IoT Integration:** Model integration into the existing TypeScript platform infrastructure and AWS IoT network, architected for multi-factory scaling.
6. **Proof of Concept Evaluation Report:** Formal benchmark documentation detailing SKU accuracy, defect recall, and system latency against the four primary KPIs following Month 3.
7. **Multilingual Project Documentation & Manuals:** Comprehensive technical design documents and operational user manuals delivered in both Thai and English

4. Timeline & Milestones

Phase 1: Proof of Concept (Months 1-3)

Period	Key Activities	Milestone
Month 1	Install industrial cameras on the production line. Record 7 days of video footage. Review footage to validate camera placement and coverage.	Camera installation complete 7-day reference video set collected
Month 2	Develop models for SKU classification and defect detection. Implement production logic (idle-stop after 10s, auto SKU-switch detection) and initial training and testing.	First working model version (classification + defect detection)

Period	Key Activities	Milestone
Month 3	Tune and validate models against target accuracy. Test edge cases (e.g., bottom-of-bag defects). Finalize processing approach (queue-based / real-time / cloud) based online speed constraints.	PoC complete ≥ 98% accuracy for SKU classification and severe defect detection

Phase 2: Production Integration & Rollout (Months 4-10)

Period	Key Activities	Milestone
Month 4	Finalize system architecture for production (queue-based / real-time / cloud) and integration plan with the existing platform.	Architecture decision approved
Month 5	Integrate trained models into the existing platform (TypeScript). Connect to AWS IoT for camera/data flow.	Model integrated into existing system
Month 6	Build the SKU/artwork upload feature so users can register new bag designs without code changes.	New-SKU upload features complete
Month 7	Internal testing across dust/high-speed conditions. Performance and latency tuning (≤ 1-minute end-to-end latency).	System passes internal performance testing
Month 8	Pilot deployment at 1-2 factories. Monitor accuracy and stability under living conditions.	Pilots live at 1-2 factories
Month 9	Roll out to remaining factories.	System lives at all factories
Month 10	User Acceptance Testing (UAT). Prepare and deliver User Manual and project documentation.	UAT passed Documentation delivered Project handover complete

5. Out of Scope

- Detection of minor packaging defects (e.g., small pinholes) that do not affect product integrity.
- Hardware procurement and physical installation beyond the cameras required for the pilot and rollout.